



Poster Abstract: Enhancing Fault Resilience of Air Quality Monitoring in San Joaquin Valley: A Data Equity Analysis

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ABSTRACT

This paper examines fault resilience among citizen-science air quality monitoring networks in California's economically challenged San Joaquin Valley (SJV). We examine disparities in monitoring capabilities and data equity between the SJV and the San Francisco Bay Area. We found significant inequities through experimental analysis simulating sensor failures. Our results emphasize the need for reliable monitoring systems and advanced modeling algorithms in resource-limited areas.

KEYWORDS

Air Quality, Fault Resilience, Data Equity

ACM Reference Format:

Zhizhang Hu*, Shangjie Du*, Yuning Chen, Xuan Zhang, Wan Du, Asa Bradman, and Shijia Pan. 2023. Poster Abstract: Enhancing Fault Resilience of Air Quality Monitoring in San Joaquin Valley: A Data Equity Analysis. In *The 21st ACM Conference on Embedded Networked Sensor Systems (SenSys '23)*, November 12–17, 2023, Istanbul, Türkiye. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3625687.3628384>

1 INTRODUCTION

Air quality monitoring is critical for public health, especially in regions where exposure to pollutants is high. The lack of adequate monitoring and mitigation measures in regions that lack resources can exacerbate the negative impacts of poor air quality. California's San Joaquin Valley (SJV) is a region with serious economic hardships, some of the worst air quality in the United States [1], and sparse air monitoring. The SJV's distinct geography, which is encircled by mountains, serves as a basin that traps pollutants from traffic, local industry, extensive agriculture, and frequent wildfires. The polluted air results in a significant public health threat. For example, the childhood asthma rate is 15.8%, among the highest in the state [3].

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SenSys '23, November 12–17, 2023, Istanbul, Türkiye

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ACM ISBN 979-8-4007-0414-7/23/11.

<https://doi.org/10.1145/3625687.3628384>

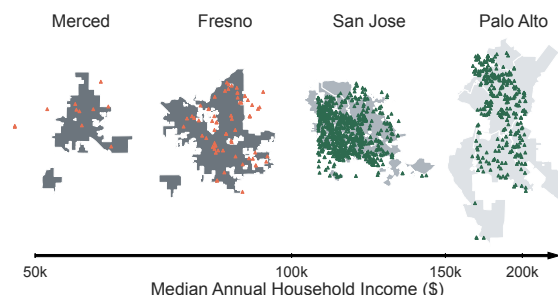


Figure 1: Purple Air sensor network deployment. Merced and Fresno are in the California Central Valley. The map's color indicates the level of median annual household income¹.

A fine-grained pollutant monitoring system with high fault resilience is crucial to better characterize air quality and protect public health in regions that lack resources, ensuring data equity to better understand environmental health threats. Fine-grained monitoring can be used to identify localized pollution spikes given the high spatial variability of pollutant distribution [2], thereby fostering informed public health decision-making. Meanwhile, high fault resilience is vital in maintaining reliable monitoring in areas where the impact of system failures is significant due to limited resources.

However, the Central Valley exhibits disparities in both fine-grained monitoring and fault resilience. For instance, the Central Valley has a notably lower density of regulatory and citizen-science PM_{2.5} sensor deployment compared to more affluent regions, indicating a gap in equitable access to environmental data. For example, Figure 1 illustrates locations of Purple Air² sensors in four cities. Merced and Fresno, two cities in the Central Valley, have a significantly lower number of sensors than the other two cities in the San Francisco Bay Area. This limited deployment results in considerable monitoring gaps, which potentially affects the availability of essential information for equitable public health interventions.

Additionally, the sparse deployment compromises the fault resilience of the monitoring systems. For the SJV, any failed sensor can leave communities uninformed and vulnerable for extended periods. This vulnerability is further exacerbated by the challenges

²One of the US nationwide air quality monitoring sensor networks: map.purpleair.com

associated with imputing missing data through data-driven methods in such areas. The widely-spaced sensors make the imputation process more challenging and less reliable.

In this paper, we characterize and compare the fault resilience of air quality monitoring networks with different deployment granularity in California's San Joaquin Valley and the San Francisco Bay Area. We conduct a sensor imputation experiment to measure and compare sensor networks' fault resilience in the aforementioned four cities, by mimicking random sensor failures of different ratios. Our findings suggest a significant inequity of fault resilience between regions with different access to air monitoring resources.

2 EXPERIMENTAL ANALYSIS

We conduct an experimental analysis of missing sensor imputation to assess the sensor network's fault resilience. We collect hourly PM2.5 data from Purple Air monitors between 08/22 and 08/30 in 2023. With an increasing ratio of failed sensors, the degradation of model performance reveals the system's resilience in monitoring regional air quality with a sparse distribution of sensors.

We first randomly select one sensor as the prediction target. We then train a multivariate regression model with data from its nearby sensors as the baseline, which simulates there is no sensor loss in the network. Then, a proportion (25%, 50%, 75%) of nearby sensors are randomly selected as the failed sensors with a constant reading. For each proportion, a new model is trained on this partially-corrupted dataset. The system's fault resilience capability is quantified by the model degradation. We adopt two settings for selecting nearby sensors: **(A). sensing range** and **(B). sensor distance**. In setting A, we fix the radius of selection, and all sensors within the circle centered on the prediction target are considered nearby sensors. We select a radius of seven miles for the experiment, given it is the shortest distance that enables experiments in four cities. In setting B, we use the target sensor's eight closest sensors as nearby sensors given there are at most nine sensors in Merced. We use the Coefficient of Variation of Root-Mean Squared Error (CVRMSE) as the evaluation metric, which shows the average amount that a model's predictions differ from the real data. The CVRMSE is defined as:

$$\text{CVRMSE} = \frac{1}{\bar{Y}} \sqrt{\frac{\sum_{i=1}^N (\tilde{Y}_i - Y_i)^2}{N}}, \quad (1)$$

where Y is the ground truth, $\bar{Y}$ is the average of ground truth, $\tilde{Y}$ is prediction, N is the number of samples. A lower CVRMSE indicates a better model prediction. We repeat experiments 10 times for each city and report the mean and standard deviation as results.

Figure 2 presents the results of the two settings. In Figure 2 (a), models of the Central Valley achieve a CVRMSE as low as 20% in the absence of sensor loss, while models of the Bay Area yield results approaching 0. This difference is the result of the varying density of sensors within the same radius. When 75% of nearby sensors are missing, Bay Area models' performance surpasses that of the SJV models without sensor loss. In Figure 2 (b), with the fixed number of nearby sensors, SJV models exhibit a rapid degradation in performance and high prediction variance. The high degradation rate underscores the system's low fault resilience in the SJV, implying

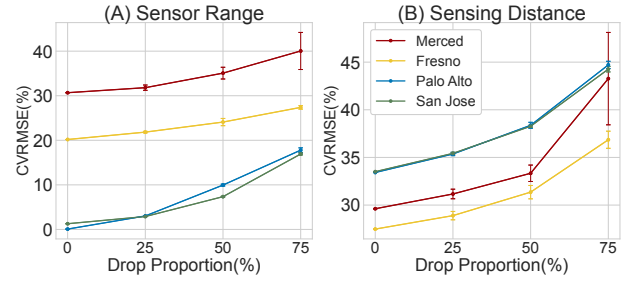


Figure 2: (A) Nearby sensors are defined within a circle of radius 7 miles; (B) Nearby sensors are the top 8 closest.

that the impact of losing one sensor is substantial to the monitoring system. The elevated variance indicates that the model's predictions become less reliable when nearby sensors are absent. These findings highlight a clear disparity in fault resilience between systems in the two regions. Also, these results emphasize the need for enhanced monitoring resilience in areas with limited sensors, such as the SJV, to ensure equitable and dependable air quality monitoring.

3 CONCLUSION AND DISCUSSION

We characterize and compare the air quality monitoring sensor network's fault resilience between two regions with different spatial sensing granularity and consequent data equity. Our results show critical data and fault resilience inequity between the lower-income California San Joaquin Valley (SJV) and higher-income San Francisco Bay Area. Given the SJV's unique geology, in future work will evaluate data-driven air quality imputation models with multiple information inputs. A model that aggregates spatial-temporal information such as topography, land usage (residence/industry/agriculture, etc.), traffic volume, wind direction, and other factors will likely increase the accuracy of fine-grained forecasting with high fault tolerance. Considering the unique pollution challenges in the SJV, we will also examine predictive distribution models of PM2.5 and other air pollutants, comparing regulator-grade monitoring with lower-cost sensors.

ACKNOWLEDGEMENT

This research is supported by a UC Merced Spring 2023 Climate Action Seed Competition Grant.

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²Data source: <https://www.census.gov/>